

Microprocessors Programming & Interfacing-Tutorial 2

Introduction to Microprocessors & 8086

Q1.

Suppose $DS=A090_H$, $DI=2000_H$, $SI=0302_H$, $BX=1000_H$, $SS=2300_H$,
 $BP=3200_H$

Determine the possible memory locations that can be accessed using the instructions given below.

MOV AX, SS:[DI]

MOV AX, [BX+SI+10_H]

MOV [BP], DL

<u>CS</u>	<u>IP</u>	<u>Instruction Address</u>
SS	SP (or) BP	Stack address
DS	BX,DI,SI an 8-bit number 16 - bit number	Data address
ES	DI for string Instructions	String destination address

Q1.

For 8086 processor, Suppose DS=A090_H, DI=2000_H, SI=0302_H, BX=1000_H, SS=2300_H, BP=3200_H

Determine the memory locations that are accessed using the instructions given below.

MOV AX, SS:[DI] → 25000_H ..

MOV AX, [BX+SI+10_H] → A1C12_H..

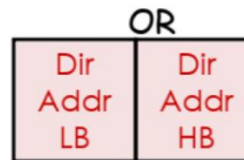
MOV [BP], DL → 26200_H

Q2. Determine addressing mode and machine code for the following MOV [SI],CL



D = 0 (Direction from Reg)
 = 1 (Direction to Reg)

W = 0 (Data - byte)
 = 1 (Data - word)



EAX/AX/AL	000
EBX/BX/BL	011
ECX/CX/CL	001
EDX/DX/DL	010
ESP/SP/AH	100
EBP/BP/CH	101
ESI/SI/DH	110
EDI/DI/BH	111

REG

MOD	00	01	10	11		
R/M					W = 0	W = 1
000	[BX] + [SI]	[BX]+[SI]+d8	[BX]+[SI]+d16		AL	AX
001	[BX] + [DI]	[BX]+[DI]+d8	[BX]+[DI]+d16		CL	CX
010	[BP] + [SI]	[BP]+[SI]+d8	[BP]+[SI]+d16		DL	DX
011	[BP] + [DI]	[BP]+[DI]+d8	[BP]+[DI]+d16		BL	BX
100	[SI]	[SI]+d8	[SI]+d16		AH	SP
101	[DI]	[DI]+d8	[DI]+d16		CH	BP
110	d16	[BP] + d8	[BP] + d16		DH	SI
111	[BX]	[BX]+d8	[BX]+d16		BH	DI

- Indirect addressing
- 100010,0,0,00,001,100
- 880C

Q3. Assume microprocessor acting in **16 bit mode of operation**. For the instructions: **MOV [SI + 100H], EAX**. Determine the addressing mode and the machine code

Ans. **Register relative addressing mode**

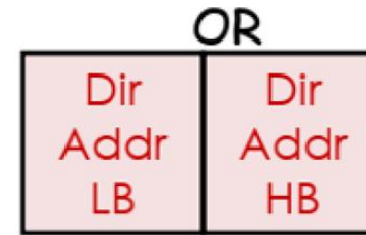
MOV [SI + 100H], EAX

100010 01 10 000 100

66 8984 00 01h

D = 0 (Direction from Reg)
 = 1 (Direction to Reg)

W = 0 (Data - byte)
 = 1 (Data - word)



- ▶▶ 16-bit mode – 32 bit addressing mode 67_H
- ▶▶ 32-bit mode – 16 bit address mode 67_H

- ▶▶ 16-bit mode – 32 bit register 66_H
- ▶▶ 32-bit mode – 16 bit register 66_H

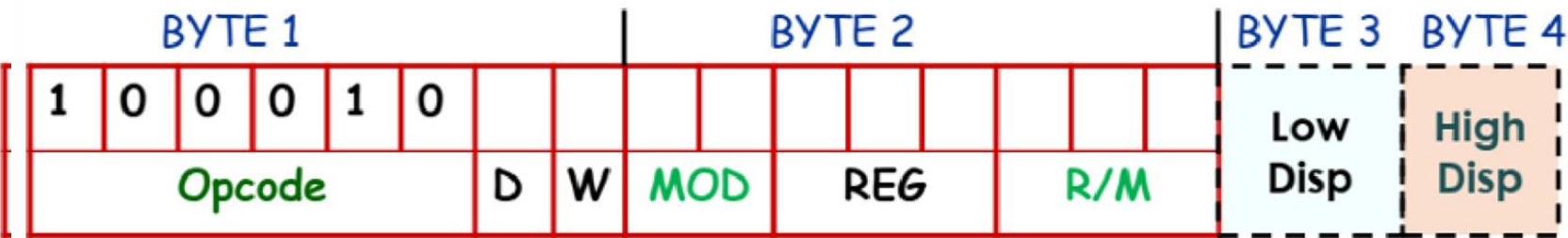
MOD	00	01	10	11	
R/M					
					W = 0 W = 1
000	[BX] + [SI]	[BX]+[SI]+d8	[BX]+[SI]+d16		AL AX
001	[BX] + [DI]	[BX]+[DI]+d8	[BX]+[DI]+d16		CL CX
010	[BP] + [SI]	[BP]+[SI]+d8	[BP]+[SI]+d16		DL DX
011	[BP] + [DI]	[BP]+[DI]+d8	[BP]+[DI]+d16		BL BX
100	[SI]	[SI]+d8	[SI]+d16		AH SP
101	[DI]	[DI]+d8	[DI]+d16		CH BP
110	d16	[BP] + d8	[BP] + d16		DH SI
111	[BX]	[BX]+d8	[BX]+d16		BH DI

MOD	00	01	10	11	
R/M					
					W = 0 W = 1
000	EAX	EAX+d8	EAX+d32		AL EAX
001	ECX	ECX+d8	ECX+d32		CL ECX
010	EDX	EDX+d8	EDX+d32		DL EDX
011	EBX	EBX+d8	EBX+d32		BL EBX
100	Scaled Index	Scaled Index +d8	Scaled Index +d32		AH ESP
101	d32	EBP+d8	EBP+d32		CH EBP
110	ESI	ESI+d8	ESI+d32		DH ESI
111	EDI	EDI+d8	EDI+d32		BH EDI

EAX/AX/AL	000
EBX/BX/BL	011
ECX/CX/CL	001
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EDI/DI/BH	111

Q3. Mention the assembly instruction for the X86 Machine Code mentioned below. It is given that the processor is 80386 which is working in 16-bit operation mode.

a) MOV EDI, [EBX+ 673948H]



MOV [SI + 100H], EAX

100010 01 10 000 100

OR



66 8984 00 01h

D = 0 (Direction from Reg)
= 1 (Direction to Reg)

W = 0 (Data - byte)
= 1 (Data - word)

this will not be considered in the best 9 labs.

▶▶ 32-bit mode – 16 bit address mode 67_H

▶▶ 16-bit mode – 32 bit register 66_H

▶▶ 32-bit mode – 16 bit register 66_H

MOD	00	01	10	11		
R/M					W = 0	W = 1
000	[BX] + [SI]	[BX]+[SI]+d8	[BX]+[SI]+d16		AL	AX
001	[BX] + [DI]	[BX]+[DI]+d8	[BX]+[DI]+d16		CL	CX
010	[BP] + [SI]	[BP]+[SI]+d8	[BP]+[SI]+d16		DL	DX
011	[BP] + [DI]	[BP]+[DI]+d8	[BP]+[DI]+d16		BL	BX
100	[SI]	[SI]+d8	[SI]+d16		AH	SP
101	[DI]	[DI]+d8	[DI]+d16		CH	BP
110	d16	[BP] + d8	[BP] + d16		DH	SI
111	[BX]	[BX]+d8	[BX]+d16		BH	DI

MOD	00	01	10	11		
R/M					W = 0	W = 1
000	EAX	EAX+d8	EAX+d32		AL	EAX
001	ECX	ECX+d8	ECX+d32		CL	ECX
010	EDX	EDX+d8	EDX+d32		DL	EDX
011	EBX	EBX+d8	EBX+d32		BL	EBX
100	Scaled Index	Scaled Index +d8	Scaled Index +d32		AH	ESP
101	d32	EBP+d8	EBP+d32		CH	EBP
110	ESI	ESI+d8	ESI+d32		DH	ESI
111	EDI	EDI+d8	EDI+d32		BH	EDI

EAX/AX/AL	000
EBX/BX/BL	011
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